

SQL Server Backup & Restore Operations Runbook

Version 1.0 | Production operations template

Purpose. Provide a repeatable, reviewable workflow for SQL Server backup and recovery operations while separating inspection from high-impact execution.

SAFETY: Restore and recovery commands can overwrite data, change database state, or extend an outage. Never execute a template until the database, target, backup chain, paths, recovery objective, and approvals have been verified. Test in non-production.

1. Scope

Covers routine FULL/DIFF/LOG backup operations, backup history review, restore planning, restore to a new database, point-in-time recovery planning, tail-log decision guidance, and post-restore validation.

2. Operating principles

- Separate diagnostics from change execution.
- Know the recovery objective before choosing commands.
- Build and peer-review the complete restore sequence before starting.
- Prefer a separate target database for validation when practical.
- Do not assume a successful backup is recoverable; perform restore testing.
- Treat backup files and encryption material as sensitive production data.

3. Routine backup procedure

1. Confirm database, environment, backup type, recovery model, and policy.
2. Confirm destination capacity, access, retention, encryption, and maintenance conflicts.
3. Run the read-only inventory/history scripts and review recent backup state.
4. Customize the appropriate FULL, DIFF, or LOG template. Review the generated command.
5. Obtain required approval; only then remove the safety stop by setting @Approved = 1.
6. Execute, capture output, confirm msdb history, and retain operational evidence.
7. Perform scheduled restore validation according to organizational policy.

4. Restore / recovery decision flow

Question	Decision
What is the objective?	Test restore, recovery from accidental change, corruption recovery, migration, or DR.
What recovery point is required?	Latest possible, end of a backup, or exact point in time.
What backups are required?	Validated FULL, optional matching DIFF, then required LOG chain in order.
Can you restore separately?	Prefer a new target for validation when operationally possible.
Need more logs applied?	Use NORECOVERY until the final intended restore.
Final step?	Use RECOVERY only when no additional backups are intended.

5. Point-in-time recovery

For FULL/BULK_LOGGED recovery workflows, identify a valid base backup and the required log chain. Confirm the exact recovery timestamp and time zone with the incident owner. Apply FULL (and a matching DIFF if selected) WITH NORECOVERY, then log backups in LSN order. On the final applicable log restore, use STOPAT and RECOVERY. Never skip a required log backup.

6. Tail-log backup

A tail-log backup may preserve log records not yet captured in a log backup before certain restore scenarios. Whether it is possible or appropriate depends on database state and the incident. The supplied template does not automatically add NORECOVERY or NO_TRUNCATE. Make that choice only from the actual recovery plan.

7. Restore-to-new-database procedure

1. Run RESTORE HEADERONLY and FILELISTONLY against the trusted backup.
2. Confirm target server/version, target database name, file paths, free capacity, and encryption prerequisites.
3. Ensure the target database name does not already exist.
4. Map every logical data/log file to an approved physical target path. Multi-file databases require additional MOVE clauses.
5. Peer-review the RESTORE command and execute under the approved change/incident.
6. Verify ONLINE state, restore history, ownership, users/logins, permissions, jobs/integrations, and application connectivity.
7. Run DBCC CHECKDB in an approved window and obtain application/business validation.

8. Common stop conditions

- Target database or file path is ambiguous.
- Required backup or log in the chain is missing.
- Recovery timestamp/time zone is not confirmed.
- Encryption key/certificate required for restore is unavailable.
- Backup header/file list does not match the expected database.
- Insufficient target storage or unclear application impact.
- Operator is considering WITH REPLACE without an explicit, reviewed overwrite plan.

9. Evidence to capture

Ticket/change ID, operator/reviewer, source and target, backup files used, backup/restore timestamps, recovery point, generated commands, SQL output/errors, post-restore validation, integrity-check result, and application/business sign-off.

10. Disclaimer

This runbook is a general operational template, not a substitute for your organization's tested disaster-recovery plan or Microsoft product documentation. Validate every procedure against your SQL Server version, architecture, security controls, and recovery requirements.